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Shock tube and modeling study of ignition delay times of propane under $O_2/CO_2/Ar$ atmosphere

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ABSTRACT

Pressurized oxy-fuel combustion is considered to be a new generation of oxy-fuel combustion technology owing to its low emission and high efficiency. In this study, the ignition delay times (IDTs) for propane under $O_2/CO_2/Ar$ atmospheres were measured in a shock tube at varying pressures and equivalence ratios. A chemical kinetic model (OXYMECH 2.0) for pressurized oxy-fuel combustion was developed by updating several key elementary reactions in our previous model (OXYMECH 1.0). OXYMECH 2.0 was validated based on experimentally measured IDTs for methane, ethane, and propane in O_2/CO_2, O_2/N_2, and O_2/Ar atmospheres in a pressure range of 1–250 atm, as well as laminar flame speeds (LFSs) in pressure and temperature ranges of 1–4 atm and 298–543 K, respectively. The OXYMECH 1.0, Aramco 3.0, CRECK, and DTU models were also evaluated; the results indicate that OXYMECH 2.0 performs better than other models in terms of predicting the IDTs and LFSs of C_1–C_3 alkanes in O_2/CO_2 and O_2/N_2 atmospheres. A detailed comparison between the OXYMECH 2.0 and Aramco 3.0 models was also performed. The influences of the pressure, equivalence ratio, and CO_2 concentration on the IDTs of propane were analyzed in detail.

1. Introduction

Oxy-fuel combustion is a possible solution for reducing the emission of greenhouse gases, as CO_2 capture can easily be accomplished from a concentrated CO_2 product stream [1,2]. The high CO_2 concentrations in oxy-fuel atmospheres can change the combustion process significantly compared with conventional combustion [3]. Moreover, pressurized oxy-fuel combustion is considered a new generation of oxy-fuel combustion technology owing to its low emissions and high efficiency [4]. An Allam cycle is any supercritical CO_2 Brayton cycle that utilizes oxy-fuel and is direct-fired with energy generated in a high-pressure turbine; a 50 MWh natural gas-fired Allam cycle plant is being demonstrated in La Porte, Texas [5]. Therefore, investigations of the ignition characteristics and chemical kinetics of fuels under O_2/CO_2 atmospheres are required.

Shock tubes have been a significant part of combustion studies and integral sources of data that have been used to characterize the ignition properties of real fuels [6,7]. The ignition delay time (IDT), typically measured with shock tubes, is a key parameter in the development and validation of chemical kinetics models for fuels. There have been numerous shock tube investigations focusing on hydrogen [7], methane [6–11], and ethane [4] under O_2/CO_2 atmospheres. For example, Shao et al. [7] measured the IDT of methane under an O_2/CO_2 atmosphere at high pressures of up to 250 atm in a shock tube. As it is a key fuel for characterizing the oxidation kinetics from light to heavy hydrocarbons [12] and a major component of liquefied propane gas (LPG), the ignition characteristics of propane have been investigated extensively under O_2/N_2 atmospheres [13-15] and O_2/Ar atmospheres [16,17]. However, shock tube experiments investigating propane under O_2/CO_2 atmospheres are still lacking.

Cai et al. [9] investigated the discrimination of the GRI 3.0 [18], Aramco 2.0 [19], and ITV [20] models for oxy-methane combustion using IDTs measured by Koroglu et al. [21] at 0.75 and 3.7 atm; the results indicated that none of the three models showed a predictive advantage over the other two models. Shao et al. [7] found that the Aramco 2.0 model could predict their experimental results more accurately than the FFCM-1 model could [22]. Moreover, Liu et al. [10] compared the performance of the FFCM-1, Aramco 1.3 [19], CRECK [23], USC 2.0 [24], and GRI

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3.0 models using IDTs obtained experimentally at 0.8 atm and reported by Hargis and Petersen [8] at 1.75 and 10 atm under O_2/CO_2 and O_2/N_2 atmospheres. They found that none of these models showed good agreement with the experimental data under the O_2/CO_2 atmosphere; an oxy-fuel combustion model (named OXYMECH 1.0 [25]) was then proposed based on the Aramco and FFCM-1 models. OXYMECH 1.0 has been validated using numerous IDTs for ethane and methane under O_2/CO_2, O_2/N_2, and O_2/Ar atmospheres. However, an oxy-fuel combustion reaction kinetic model needs to be developed for CH_4 and C_2H_6 to C_3H_8 over a pressures range of 1–250 atm.

Recently, Hashemi et al. [26] investigated the high-pressure oxidation of propane using a laminar flow reactor and established a brand new chemical kinetic model (DTU). The predictions obtained using the DTU model were in satisfactory agreement with the tested species profiles as well as the IDTs and LFSs. However, the model still needs to be validated based on the IDTs of methane, ethane, and propane under O_2/CO_2 atmospheres.

A reliable oxy-fuel combustion model needs to be validated using IDTs, LFSs, and species profiles. The LFSs of methane and ethane under O_2/CO_2 atmospheres have been measured by many researchers [27–30]. Therefore, the oxy-fuel combustion reaction kinetic model in this study is validated using the LFSs of methane and ethane under O_2/CO_2 and O_2/N_2 atmospheres.

The H-atom abstraction reactions of propane by OH and HO_2 exhibit relatively high sensitivity to the IDT at high pressure [31]. Sivaramakrishnan et al. [32] measured the rate constants for the C_3H_8 + OH <=> C_3H_7 + H_2 reaction at high temperatures using the reflected shock tube technique with multi-pass absorption spectrometric detection of OH. Hashemi et al. [26] calculated the rate constants of C_3H_8 + HO_2 <=> C_3H_7 + H_2O theoretically. However, the physical and chemical properties of CO_2 are considerably different from those of N_2. CO_2 is not chemically inert, and the reaction rates of three-body reactions are enhanced under O_2/CO_2 atmospheres owing to the chaperon effect of CO_2. Hu et al. [33] studied the chemical effects of CO_2 on the LFSs of oxy-methane mixtures at various oxygen concentrations and identified the main three-body reactions, including CH_3 + OH (+M) <=> CH_3OH (+M). Choudhary et al. [34] measured the reaction rate of H + O_2 (+M) <=> HO_2 (+M) near the low-pressure limit for three bath gases (Ar, N_2, and CO_2) at high temperature. Shao et al. [35] investigated the rate constants for H + O_2 (+M) <=> HO_2 (+M) at elevated pressures from 12 to 33 atm with four different collision partners (Ar, N_2, H_2O, and CO_2). Recently, Yang et al. reevaluated the reaction rate of H + O_2 (+M) <=> HO_2 (+M) with M = Ar and N_2 at elevated pressures from 1 to 150 bar. Evidently, the updating of these reactions is expected to improve the prediction performance of OXYMECH.

In this study, the IDTs of propane under $O_2/CO_2/Ar$ atmospheres were measured in a shock tube at pressures of 1.0 and 10.0 bar; equivalence ratios of 0.5, 1.0, and 1.5; high CO_2 concentrations of 60% and 80%; and temperatures in the range of 1200–1600 K. Argon was used to compensate for the changes in CO_2 content. Moreover, an updated oxy-fuel combustion reaction kinetic model, named OXYMECH 2.0, based on OXYMECH 1.0, was proposed. OXYMECH 2.0 was validated using the IDTs of propane in O_2/CO_2 atmospheres measured in the present study and in O_2/Ar atmospheres; the IDTs of CH_4 highly diluted in CO_2 at high pressures up to 250 atm; and the LFSs of methane, ethane, and propane in O_2/CO_2, O_2/N_2, and O_2/Ar atmospheres from the literature. Four chemical kinetic models (Aramco 3.0 [36], CRECK, DTU, and OXYMECH 2.0) were evaluated in detail using the IDTs and LFSs. The influence of the pressure, equivalence ratio, and CO_2 concentration on the IDTs of propane was also analyzed in detail.

Table 1
Composition of the experimental mixtures and test conditions.

Mixture	C_3H_8	O_2	CO_2	N_2	Ar	Φ	Pressure (bar)
mix-1	0.01	0.1	0.6	0	0.29	0.5	1.0, 10.0
mix-2	0.02	0.1	0.6	0	0.28	1.0	1.0, 10.0
mix-3	0.03	0.1	0.6	0	0.27	1.5	1.0, 10.0
mix-4	0.01	0.1	0.8	0	0.09	0.5	10.0
mix-5	0.02	0.1	0.8	0	0.08	1.0	10.0
mix-6	0.03	0.1	0.8	0	0.07	1.5	10.0
mix-7	0.02	0.1	0	0.6	0.28	1.0	1.0, 10.0

2. Methodology

2.1. Experimental facility

The IDTs in the present study were measured in a shock tube with an inner diameter of 10 cm, as reported in detail in our previous work [10]. The shock tube consists of an 8 m driven section and a 4 m driver section. The driven and driver sections of the shock tube are separated by one (at 1.0 bar) or two (at 10.0 bar) polyester terephthalate (PET) diaphragms. The single diaphragm was punctured by a built-in spring needle to produce a shock wave at 1.0 bar, while the double diaphragms were burst via sudden venting of the gas between them at 10.0 bar. Diaphragms with thicknesses of 50 μm and 125 μm were selected for the 1.0 bar and 10.0 bar tests, respectively. High-purity (99.999%) nitrogen and helium blends were used as driving gases. One sliding, oil-sealed, vane rotary vacuum pump (Oerlikon Leybold TRIVAC D40T) and two roots pumps (Oerlikon Leybold Ruvac WAU501) were used to evacuate the shock tube and the mixing tank to below 0.1 Pa. The experimental mixtures tested in this study are listed in Table 1. All the mixtures required settling for more than 12 h in a 300 L mixing tank after preparation according to Dalton's law of pressure. The purities of propane, oxygen, carbon dioxide, and argon were 99.99%, 99.999%, 99.999%, and 99.999%, respectively.

Five piezoelectric pressure transducers (PCB 111A24) were placed along the sides of the shock tube at intervals of 20 cm, to record the incident shock velocity. The pressure was captured using a piezoelectric pressure transducer (Kistler 603B1) placed 2 cm from the end wall, while a photomultiplier with a band pass filter of 307 ± 10 nm was placed 2 cm from the end wall to monitor the OH* emission. The Gaseq [37] chemical equilibrium program was employed to calculate the temperature (T_5) and pressure (P_5) behind the reflected shock waves. The uncertainty in the temperature was estimated as ±21 K, and the uncertainty in the IDTs was estimated to be less than 19%. The details of the uncertainty analysis are provided in the Supplementary material.

2.2. Ignition delay time

The IDT is defined as the time interval between the arrival of the reflected shocks and the onset of ignition, which is determined by extrapolating the maximum slope of the normalized OH* emission linearly back to the zero baseline [6,7], as shown in Fig. 1. This method was also used in our previous studies [4,10]. A large addition of CO_2 to the mixture results in a slight increase in the pressure during combustion, which is called shock bifurcation; this can make it difficult to define the arrival of the main reflected wave. In this study, we determined the arrival time of the reflected waves using the empirical expression proposed by Peterson and Hanson [38]:

$$\Delta t_{AO}(\mu s) = 4.6 M_S^{0.66} \gamma_2^{-7.1} \overline{M}^{0.57}, \tag{1}$$

where M_s is the Mach number of the incident shocks, γ_2 is the specific heat ratio upstream of the bifurcation, and $\bar{M}$ is the molec-

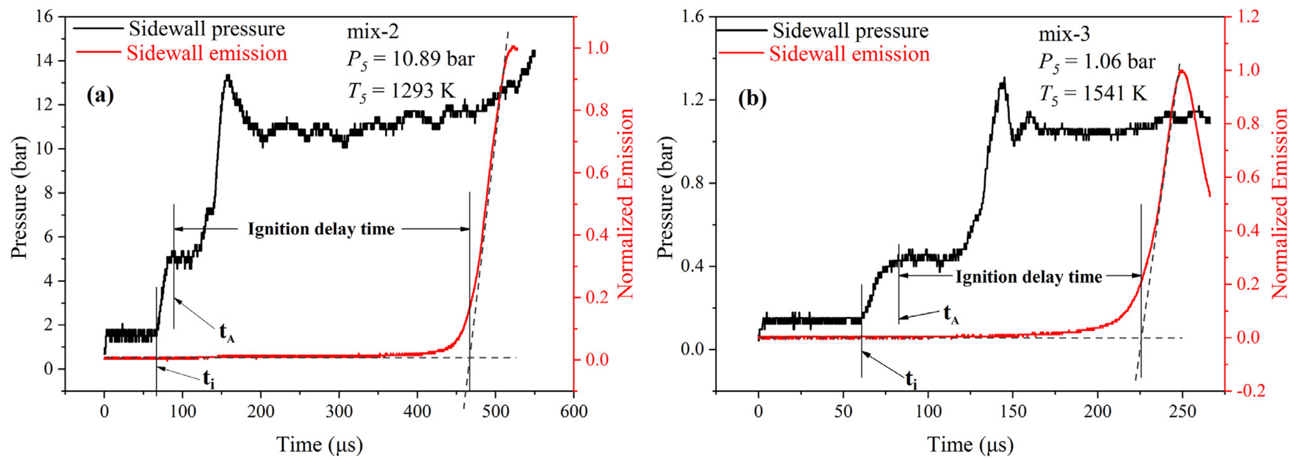

Fig. 1. Typical signals of the ignition delay time measurement: (a) for mix-2 at T = 1293 K and P = 10.89 bar; (b) for mix-3 at T = 1541 K and P = 1.06 bar.

ular weight of the mixture. The arrival time of the reflected shock waves, t_A, can be expressed as follows:

$$t_A = \Delta t_{AO} + \frac{D}{2V_R} + t_i, \tag{2}$$

where D is the diameter of the pressure transducer (Kistler 603B1), V_R denotes the reflected shock velocity, and t_i is the initial pressure rise time.

Moreover, according to Hargis and Petersen [8], reflected-shock bifurcations create eddies and a vortex sheet, which can result in the generation of increased fluid motion in the form of vorticity. This fluid motion was observed in the form of pressure oscillations, using the sidewall pressure transducers, as shown in Fig. 1. The comparisons between the calculated pressure and the measured pressure trace are presented in the Supplementary material.

2.3. Model and simulation

Table 2 lists the 17 reactions updated in OXYMECH 2.0. The important updated reactions are enabled by recent advances in both theoretical tools and experimental techniques. The H-atom abstraction reactions of the fuel molecule exhibit relatively high sensitivity to the IDT [39,40], and thus, the parameters for the hydrogen-abstraction reactions of propane with H (R332, R348) were obtained from the CRECK model. Sivaramakrishnan et al. [32] reported the rate constants for the H-atom abstraction by OH radicals from propane (R334). The hydrogen-abstraction reaction of C_3H_6 (R481) was obtained from a study conducted by Ye et al. [41]. Moreover, the chemical effects of CO_2 on the fuel combustion in O_2/CO_2 included (1) the chaperon effect of CO_2 and (2) the effects of the reactions with CO_2. Thus, the parameters for the three-body reaction of CH_3 + OH (+M) <=> CH_3OH (+M) (R2642) were obtained from GRI 3.0, and those for CH_2OH (+M) <=> H + CH_2O (+M) (R2669) were obtained from USC 2.0. The reaction of CH_2CHO + O <=> H + CH_2 + CO_2 (R2748) is important in oxy-fuel atmospheres owing to the reactant CO_2. The reaction rate constants for reaction R2748 were obtained from a report by Konnov [42]. Furthermore, reactions R147, R2585, and R2778 significantly affect the LFSs of C_1–C_3 alkanes, according to Goswami et al. [29]. The reaction rate constants for reactions R147 and R2585 were obtained from the San Diego model, while the reaction rate constants for reaction R2778 were obtained from GRI 3.0. OXYMECH 2.0 consists of 495 species and 2824 reactions, as described in the Supplementary material.

In the present study, the IDTs, rates of production (ROP), and temperature sensitivity coefficients at the ignition point were computed using the closed zero-dimensional reactor of the CHEMKIN-PRO software [47]. According to our previous study [10] and the studies of [48,49], the pressure rise resulting from non-ideal effects can be considered negligible during simulation of the relatively short IDTs of interest here. Therefore, a constant-volume, zero-dimensional chemistry model (U and V assumption) was used without modification in the current study.

3. Results and discussion

3.1. IDTs of propane under O_2/CO_2 atmosphere

Figure 2 shows the IDTs of propane diluted in different concentrations of CO_2 and measured at pressures of 1 bar and 10 bar for different equivalence ratios (Φ = 0.5, 1.0, and 1.5). In Fig. 2(a) and (b), the IDTs increase with the equivalence ratio, while the differences in the IDTs between the three equivalence ratios decrease with increasing pressure from 1 bar to 10 bar, which indicates that the effects of the equivalence ratio on the IDTs of propane under O_2/CO_2 atmospheres weaken at high pressure. A similar phenomenon was also previously observed with ethane [4]. In Fig. 2(b), the IDT simulation curves for different equivalence ratios intersect at low temperature (around 1220 K). This phenomenon might be related to pressure-dependent reactions, and the H-abstraction reactions of propane are discussed in Section 3.4. Moreover, at P = 10 bar, when the concentration of CO_2 increases from 60% to 80%, there is no statistically discernible difference between the IDTs, as shown in Fig. 2(c), (d), and (e). This phenomenon is explained in Section 3.5. We can observe that the IDTs in the CO_2 cases are slightly longer than those in the N_2 cases at P = 1 and 10 bar, as shown in Fig. 2(f). The IDTs of methane and ethane in the CO_2 cases are almost identical to those in the N_2 and Ar cases at 0.8 and 2.0 bar, and are longer than those in the N_2 cases and Ar cases at 10 bar [4,10]. It is clear that the effects of CO_2 on the IDTs of propane are stronger than those on the IDTs of methane and ethane. This phenomenon is discussed in Section 3.4.

3.2. Model evaluation

3.2.1. IDTs of propane in O_2/CO_2 atmospheres

Figure 3 shows the experimental IDTs for propane diluted in high concentrations of CO_2 (60% and 80%) with different equivalence ratios (0.5, 1.0, and 1.5) at 1 bar and 10 bar; the simulated results obtained with the OXYMECH 2.0, OXYMECH 1.0, Aramco 3.0, CRECK, and DTU models are also shown. As shown in Fig. 3, OXYMECH 1.0 over-predicts nearly all of the IDTs, which indicates

Table 2
Key elementary reactions updated in the present study[a].

Number	Reaction	A	b	E_a	Source
R332	$C_3H_8 + H <=> iC_3H_7 + H_2$	1.3E+06	2.4	4471	[43]
	Updated R332	3.2E+13	0	7900	[23]
R334	$C_3H_8 + OH <=> iC_3H_7 + H_2O$	4.67E+07	1.61	−35	[44]
	Updated R334	1.81E+05	2.437	−536	[32]
R348	$C_3H_8 + H <=> nC_3H_7 + H_2$	3.49E+05	2.69	6450	[43]
	Updated R348	1.27E+14	0	10,500	[23]
R481	$C_3H_6 <=> C_3H_5$-A + H	9.16E+74	−17.6	120,000	[22]
	Duplicate	2.98E+54	−12.3	101,200	
	Updated R481	1.08E+71	−15.91	124,860	[41]
	Duplicate	6.28E+42	−8.51	98,004	
R147	$C_2H_3 + O_2 <=> CH_2CHO + O$	1.76E+12	0.15	4205	[45]
	Updated R147	3.00E+11	0.29	11	[46]
R2568	$HO_2 + O <=> OH + O_2$	1.61E+13	0	−445	[22]
	Updated R2568	4.00E+13	0	0	[24]
R2585	$HCO + H <=> H_2 + CO$	8.48E+13	0	0	[22]
	Updated R2585	5.00E+13	0	0	[46]
R2591	$CH + H <=> C + H_2$	1.09E+14	0	0	[22]
	Updated R2591	7.90E+13	0	160	[24]
R2592	$CH + H_2 <=> H + CH_2$	1.61E+14	0	3320	[22]
	Updated R2592	1.75E+14	0	−165	[46]
R2640	$CH_3 + O <=> H + CH_2O$	5.72E+13	0	0	[22]
	Updated R2640	8.43E+13	0	0	[18]
R2642	$CH_3 + OH (+M) <=> CH_3OH (+M)$	6.21E+13	−0.018	−33	[22]
	Updated R2642	6.30E+13	0	0	[18]
R2669	$CH_2OH (+M) <=> H + CH_2O (+M)$	7.37E+10	0.811	39,580	[22]
	Updated R2669	5.40E+11	0.454	3600	[24]
R2709	$HCCO + H <=> CH_2(S) + CO$	1.32E+14	0	0	[22]
	Updated R2709	1.00E+14	0	0	[24]
R2717	$C_2H_2 + O <=> H + HCCO$	8.68E+08	1.4	2206	[22]
	Updated R2717	1.63E+07	2	1900	[24]
R2719	$C_2H_2 + OH <=> H + CH_2CO$	8.67E-01	3.566	−2370	[22]
	Updated R2719	2.18E-04	4.5	−1000	[23]
R2748	$CH_2CHO + O <=> H + CH_2 + CO_2$	1.58E+14	0	0	[22]
	Updated R2748	1.50E+14	0	0	[42]
R2778	$C_2H_4 + OH <=> C_2H_3 + H_2O$	2.14E+04	2.745	2216	[22]
	Updated R2778	2.23E+04	2.745	2216	[18]

[a] Rate constants are expressed as $k = AT^b\exp(-E_a/R_uT)$ with units of calories, mole, cm^3 and s.

that the propane sub-model needs to be updated, as OXYMECH 1.0 can accurately predict the IDTs of methane and ethane in O_2/CO_2 atmospheres. Aramco 3.0 also over-predicts nearly all of the IDTs, which is attributed to the C_1–C_2 sub-model and propane sub-model. The C_1–C_2 sub-model in the Aramco model was discussed in [10] and [4] for O_2/CO_2 atmospheres, and the propane sub-model in the Aramco model will be analyzed in Section 3.3. The DTU model over-predicts the IDTs of propane at $P = 1$ bar and $\Phi = 0.5$ and 1, as shown in Fig. 3(a) and (b). For $P = 10$ bar and 60% CO_2 in the mixtures, DTU predicts the experimental results well at $\Phi = 0.5$ and 1, while it under-predicts the IDTs at $\Phi = 1.5$ in the cases with 60% CO_2, as shown in Fig. 3(c). For $P = 10$ bar and 80% CO_2 in the mixtures, the DTU model predicts the IDTs well at $\Phi = 1$ and 1.5, but over-predicts the IDTs at $\Phi = 0.5$, as shown in Fig. 3(d)–(f). The performance of the DTU model will be further evaluated in Sections 3.2.2 and 3.2.3. Some three-body reactions, reactions with CO_2, and propane H-abstraction reactions need to be updated to improve the performance of the DTU model for atmospheres with high CO_2 concentrations. The CRECK model significantly under-predicts the IDTs at $P = 1$ bar, as shown in Fig. 3(a)–(c). However, the prediction results at 10 bar are considerably better than those at 1 bar, as shown in Fig. 3, for CO_2 concentrations of 60% and 80%.

To evaluate the performance of the five models quantitatively, the absolute relative error (ARE) value was used, which is defined as follows:

$$ARE = \left|\frac{\tau_{sim} - \tau_{\exp}}{\tau_{\exp}}\right| \times 100\% \tag{3}$$

where τ_{exp} and τ_{sim} represent the measured and calculated results for a data point, respectively. Figure 4 depicts boxplots of the AREs of the five models. Boxplots are widely utilized to illustrate data distributions. The central line and symbol in the box indicate the median and average values, respectively, while the bottom and top edges of the box indicate the upper and lower quartiles, respectively. The length of the box, called the interquartile range (IQR), reflects the concentration degree of the data. The whiskers extend to the most extreme data points, excluding outliers. As shown in Fig. 4, the average and median ARE values for OXYMECH 2.0 are the lowest among the five models. Moreover, the distribution of the ARE values for OXYMECH 2.0 is the most concentrated owing to it having the smallest IQR value. These results indicate that the performance of OXYMECH 2.0 is better than that of the other four models, and OXYMECH 2.0 is suitable for the prediction of propane ignition under O_2/CO_2 atmospheres.

3.2.2. IDTs of propane in O_2/Ar atmospheres

Lam et al. [16] measured propane IDTs in a lean mixture (0.8% C_3H_8/8% O_2/Ar) over the temperature range of 980–1400 K at nominal pressures of 6, 24, and 60 atm using a shock tube. Hu et al. [50] measured the IDTs of propane/oxygen/argon mixtures at a pressure of 20 bar in a shock tube at different equivalence ratios (0.5, 1.0, and 2.0), and temperatures (1100−1500 K). These IDT data for propane under O_2/Ar atmospheres [16,50] were also utilized to further validate the OXYMECH 2.0 model. Figure 5 shows the measured IDTs and the simulated results obtained using the OXYMECH 2.0, Aramco 3.0, and DTU models. It is noted that both the OXYMECH 2.0 and DTU models can predict the experimental results better than Aramco 3.0 can over the pressure range of 6–60

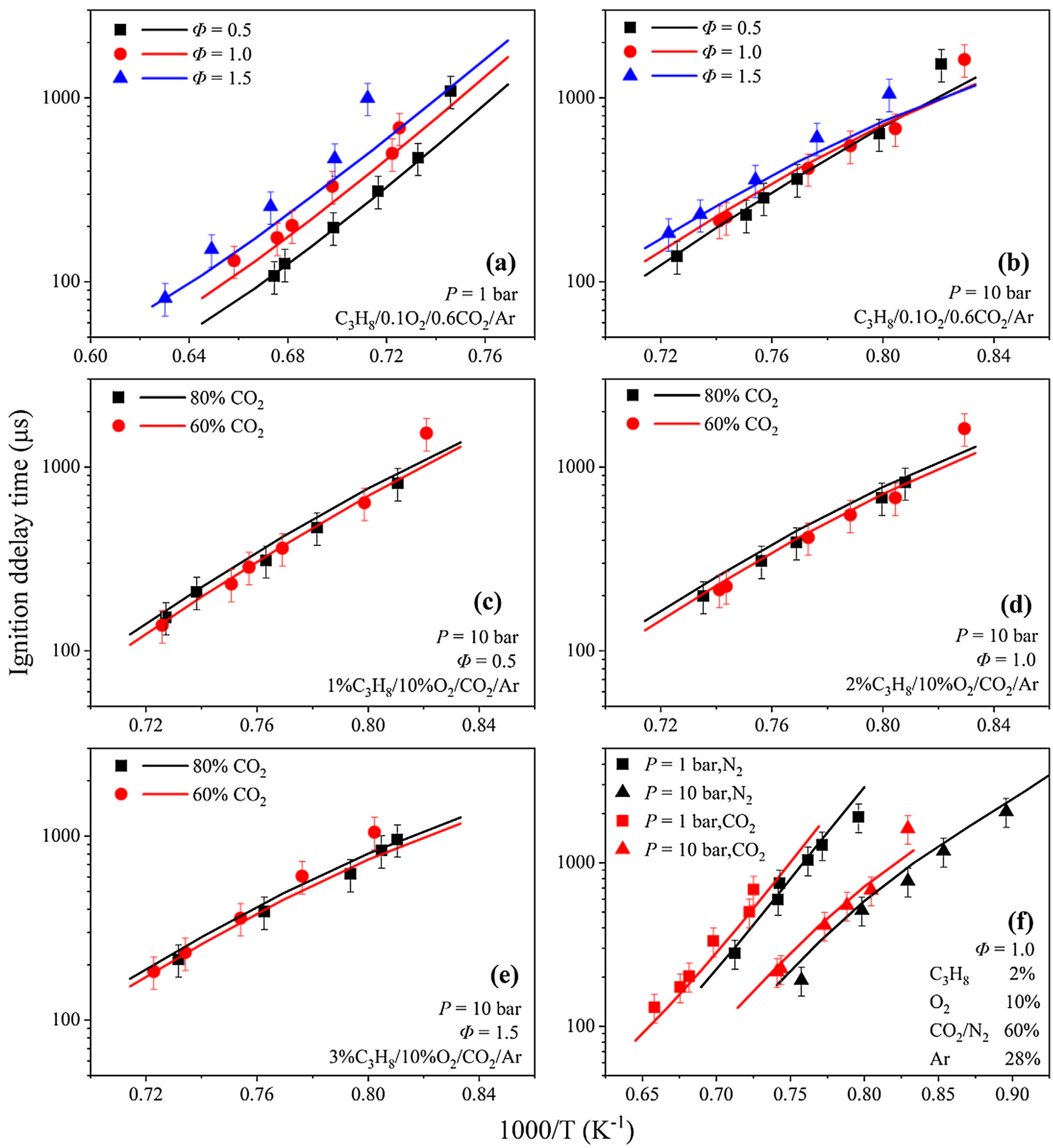

Fig. 2. IDT measurements for $C_3H_8/O_2/CO_2/Ar$ mixtures and simulation results obtained with OXYMECH 2.0.

atm. Moreover, the IDTs of propane/hydrogen blends under O_2/Ar atmospheres [51,52] were also used to validate the OXYMECH 2.0 and DTU models. The simulated and experimental results are shown in the Supplementary material. Figure 6 displays the boxplots of the AREs of the OXYMECH 2.0 and DTU models OXYMECH 2.0 clearly has slightly smaller IQRs and average AREs than the DTU model does, which indicates that the predictions obtained using OXYMECH 2.0 are slightly better than those obtained using the DTU model.

Moreover, simulations of the IDTs in our previous studies of methane [10] and ethane [4] using OXYMECH 2.0 and DTU are also presented in the Supplementary material. Figure 7 displays the boxplots for the AREs of the OXYMECH 2.0 and DTU models. The performance of the OXYMECH 2.0 model for the IDTs of methane and ethane is better than that of the DTU model owing to the smaller average errors and IQRs.

3.2.3. IDTs of methane in O_2/CO_2 atmosphere at high pressure

Shao et al. [7] measured the IDTs of CH_4 highly diluted in CO_2 at high pressures of up to 250 atm of interest in super-critical CO_2 cycles. Aramco 3.0 and DTU models were employed for comparison with OXYMECH 2.0. Figure 8 shows the experimental results and simulated results obtained with the OXYMECH 2.0, Aramco 3.0, and DTU models. As shown in Fig. 8, DTU over-predicts most of the IDTs, whereas OXYMECH 2.0 and Aramco 3.0 can both predict the data reasonably accurately. The boxplots for the AREs of the

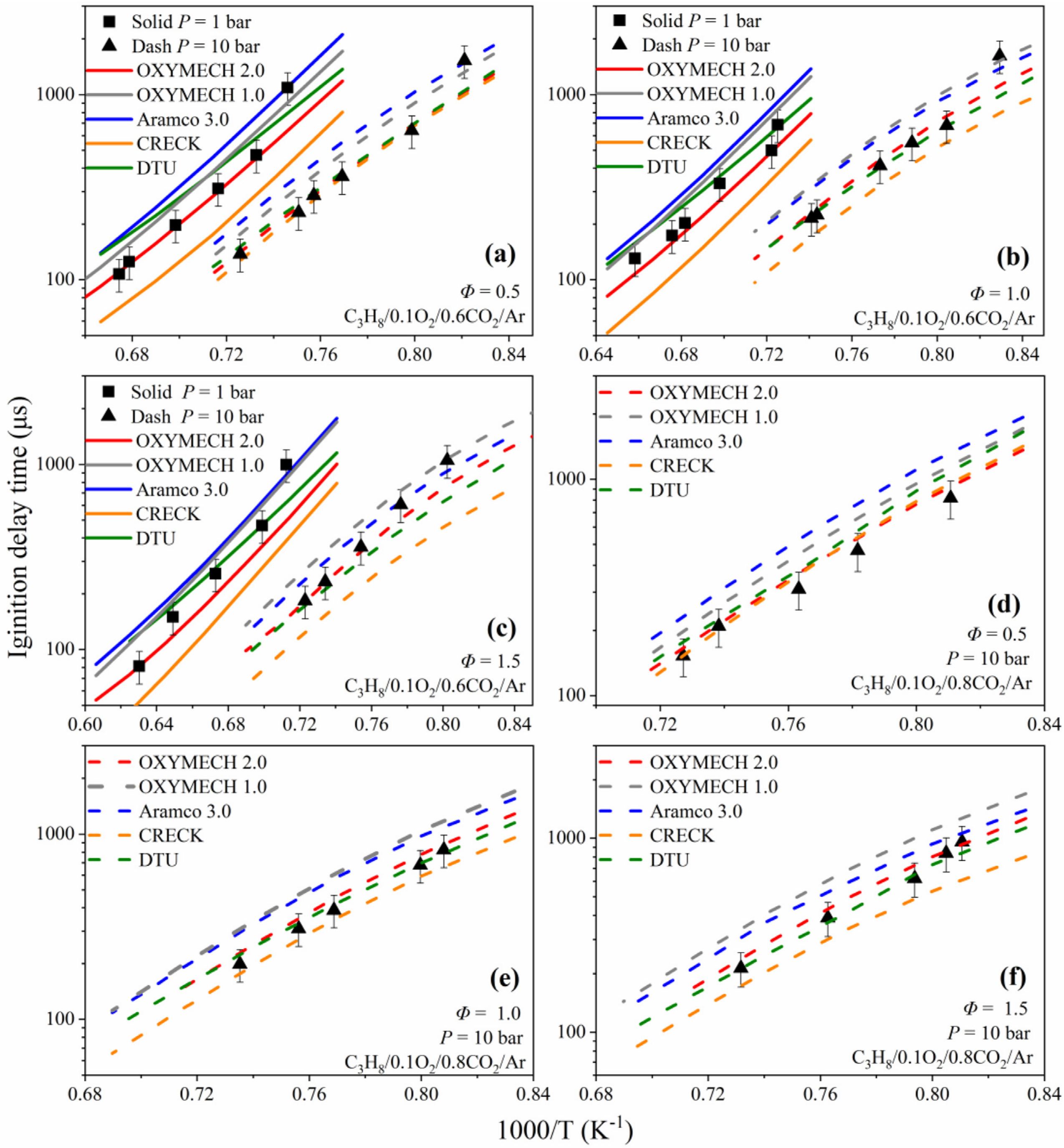

Fig. 3. Comparison of experimental and predicted IDTs in the present study simulated with five models.

OXYMECH 2.0, Aramco 3.0, and DTU models, as shown in Fig. 8, demonstrate that OXYMECH 2.0 has the smallest average and IQR values, which indicates OXYMECH 2.0 exhibits good performance for the prediction of the IDTs of CH_4 at high pressure.

3.2.4. LFSs of propane, ethane, and methane

Figure 9 shows the predictions for the LFSs of propane [53], ethane [27,29], and methane [28,30] obtained using the OXYMECH 2.0 and Aramco 3.0 models. As shown in Fig. 9(a) and (b), the LFSs of propane diluted with CO_2 and N_2 are predicted accurately by OXYMECH 2.0, while Aramco 3.0 under-predicts the LFSs at the fuel-rich side. For the LFSs of ethane in O_2/CO_2 and O_2/N_2 atmospheres, as shown in Fig. 9(c) and (d), OXYMECH 2.0 and Aramco 3.0 predict the LFSs accurately in the case of the O_2/N_2 atmosphere but slightly under-predict the LFSs at the fuel-rich side in the case of the O_2/CO_2 atmosphere. Figure 9(e) and (f) shows that OXYMECH 2.0 accurately predicts the LFSs of methane in two atmospheres, while Aramco 3.0 under-predicts the LFSs at the fuel-lean side in the O_2/CO_2 atmosphere. The other simulated and experimental LFSs for propane, ethane, and methane are shown in the Supplementary Material. The boxplots for the AREs of the OXYMECH 2.0 and Aramco 3.0 models are shown in Fig. 9(g). All these results indicate that the OXYMECH 2.0 model can provide accurate predictions for the LFSs of methane, ethane, and propane under O_2/CO_2 and O_2/N_2 atmospheres. The differences in the LFSs

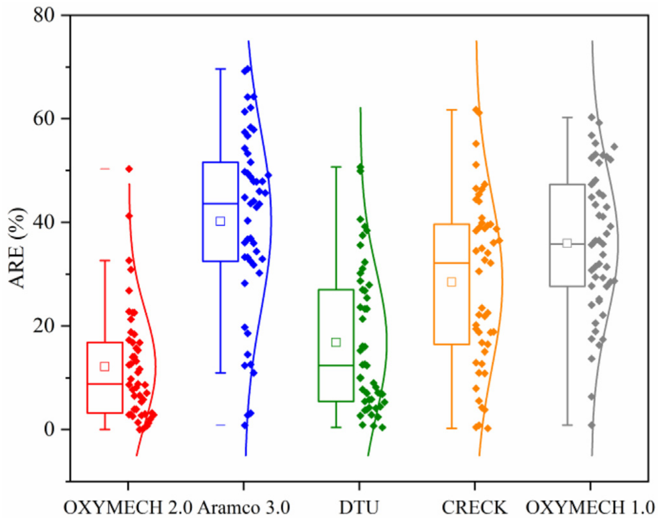

Fig. 4. Boxplots of the absolute relative errors of the five models.

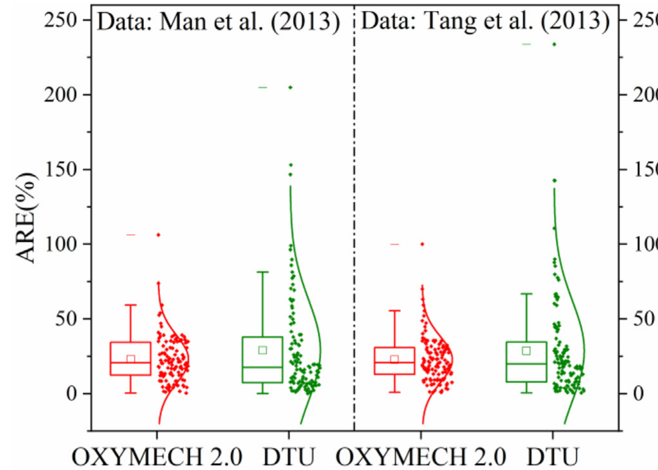

Fig. 6. Boxplots of the AREs of the OXYMECH 2.0 and DTU models for prediction of the IDTs of propane/hydrogen blends.

predicted for propane by the OXYMECH 2.0 and Aramco 3.0 models are attributed to C_3 chemistry.

The validations of the OXYMECH 2.0 model using the IDTs and LFSs of methane, ethane, and propane in O_2/CO_2 and O_2/N_2 (Ar) atmospheres presented in Sections 3.2.1–3.2.4 demonstrate that OXYMECH 2.0 is applicable for the prediction and analysis of C_1–C_3 oxy-fuel combustion.

3.3. Comparison between the OXYMECH and Aramco models

For the case of IDTs under the conditions of $\Phi = 0.5$, $P = 1$ bar, and $T = 1400$ K, a comparison of the temperature sensitivity at the ignition point between the OXYMECH 2.0 and Aramco 3.0 models was carried out. Figure 10(a) and (b) show the top 15 reactions with high sensitivity coefficients in the OXYMECH 2.0 and Aramco 3.0 models, respectively. As shown in Fig. 10, C_3H_8 + OH <=> iC_3H_7 + H_2O (R334) has the second-most negative sensitivity coefficient in OXYMECH 2.0, but this reaction is absent in Aramco 3.0. Moreover, H-atom abstraction reactions C_3H_8 + H <=> iC_3H_7 + H_2 (R332) and C_3H_8 + H <=> nC_3H_7 + H_2 (R348) have the two most negative sensitivity coefficients in Aramco 3.0, but rank 5th and 6th, respectively, in the reactions with negative sensitivity coefficients in OXYMECH 2.0.

Figure 11 displays the net reaction rates of these three reactions. Figure 11(a) shows that the rate of R334 [44] in OXYMECH 2.0 is much higher than that [36] in Aramco 3.0 due to the different reaction rate constants; this indicates that R334 consumes more C_3H_8 in OXYMECH 2.0 than in Aramco 3.0. On the other hand, the rates of both R332 and R348 [43] in OXYMECH 2.0 are lower than those [36] in Aramco 3.0 due to the different reaction rate constants, as shown in Fig. 11(b) and (c), which indicates that more C_3H_8 is converted by R332 and R348 in Aramco 3.0. Consequently, Aramco 3.0 significantly over-predicts the IDT under these conditions.

Figure 12 displays the reaction pathways at 20% fuel consumption for mix-1 at $P = 1$ bar and $T = 1400$ K in the two models. In OXYMECH 2.0, the propane is mainly dehydrogenated by OH (C_3H_8 + OH <=> iC_3H_7 + H_2O (R334) and C_3H_8 + OH <=> nC_3H_7 + H_2O (R350)), which accounts for 45.5% of the propane; in contrast, the proportion of propane dehydrogenated through reactions R334 and R350 in Aramco 3.0 is 24.2%. In Aramco 3.0, propane is mainly dehydrogenated by H (reactions R332 and R348), which accounts for 41.5% of the propane, whereas the proportion of propane dehydrogenated through reactions R332 and R348 in OXYMECH 2.0 is 22.3%. Aramco 3.0 clearly over-predicts the amount of propane dehydrogenated via reactions R332 and R348, which results in over-prediction of the IDTs.

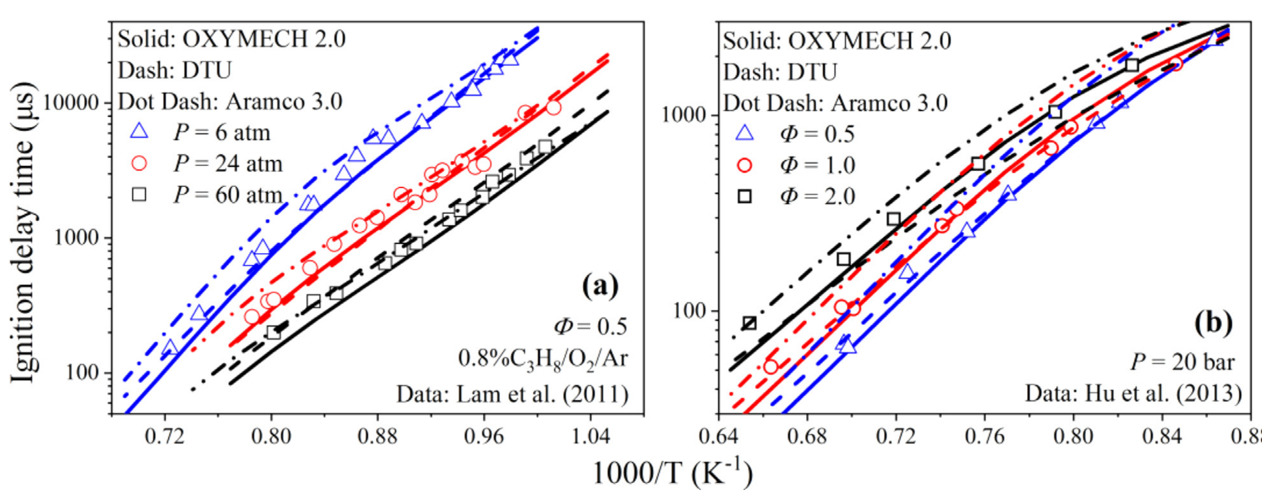

Fig. 5. Comparison between the IDTs of $C_3H_8/O_2/Ar$ obtained with the OXYMECH 2.0, DTU, and Aramco 3.0 models.

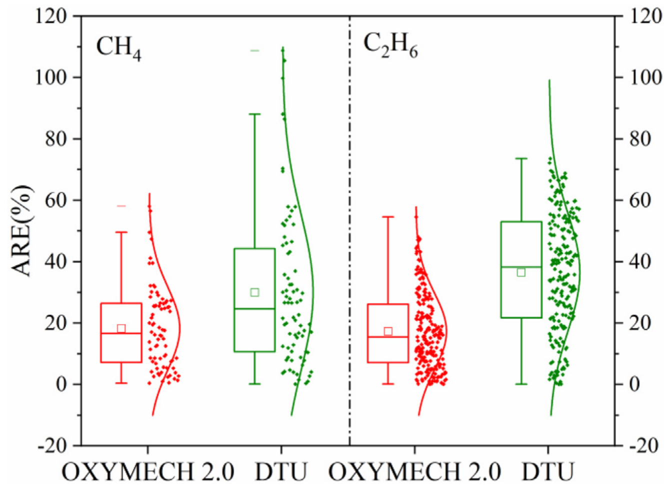

Fig. 7. Boxplots of the OXYMECH 2.0 and DTU models in our previous studies of methane and ethane.

3.4. Influence of pressure and equivalence ratio

As shown in Fig. 2(a) and (b), the IDTs of propane increase with the equivalence ratio, while the differences between the IDTs at the different equivalence ratios decrease with increasing pressure from 1 bar to 10 bar. The temperature sensitivity at the ignition point was analyzed using OXYMECH 2.0 for three equivalence ratios at P = 1 bar and P = 10 bar. Figure 13 displays the top 20 reactions with high sensitivity coefficients for propane under conditions of 60% CO_2 at T = 1450 K and P = 1 bar. As shown in Fig. 13, with the increase in the equivalence ratio from 0.5 to 1.0 and 1.5, reaction C_3H_8 + H <=> iC_3H_7 + H_2 (R332) rises from the 8th to 3rd and 1st in importance, respectively, while reaction C_3H_8 + H <=> nC_3H_7 + H_2 (R348) rises from 9th to 4th and 3rd in importance, respectively. The ignition-inhibiting reactions, R332 and R348, were thus strengthened under a higher propane concentration. Therefore, the IDTs of propane increase with the equivalence ratio at P = 1 bar and T = 1450 K. Obviously, with the increase in the equivalence ratio from the fuel-lean side to the fuel-rich side, the fuel decomposition reaction becomes more important. H-abstraction reactions of C_3H_8 by H radicals are the main reason for the increase in IDTs with the equivalence ratio at 1 atm.

Figure 14 shows the reaction pathway for propane at 20% fuel consumption, P = 1 bar, and T = 1450 K. As shown in Fig. 14(a), 8.2% of the propane is dehydrogenated to isopropyl (iC_3H_8) by H under fuel-lean conditions, whereas 13.2% of the propane is dehydrogenated to normal-propyl (nC_3H_8) by H. Under fuel-rich conditions, the proportion of propane dehydrogenated to isopropyl increases to 13.2%, while the proportion of propane dehydrogenated to normal-propyl increases to 21.1%. Therefore, the proportion of propane dehydrogenated by H (reactions R332 and R348) increases with the increase in the equivalence ratio from fuel-lean to fuel-rich, which results in longer IDTs under fuel-rich conditions compared with those under fuel-lean conditions due to the strong negative sensitivities of reactions R332 and R348.

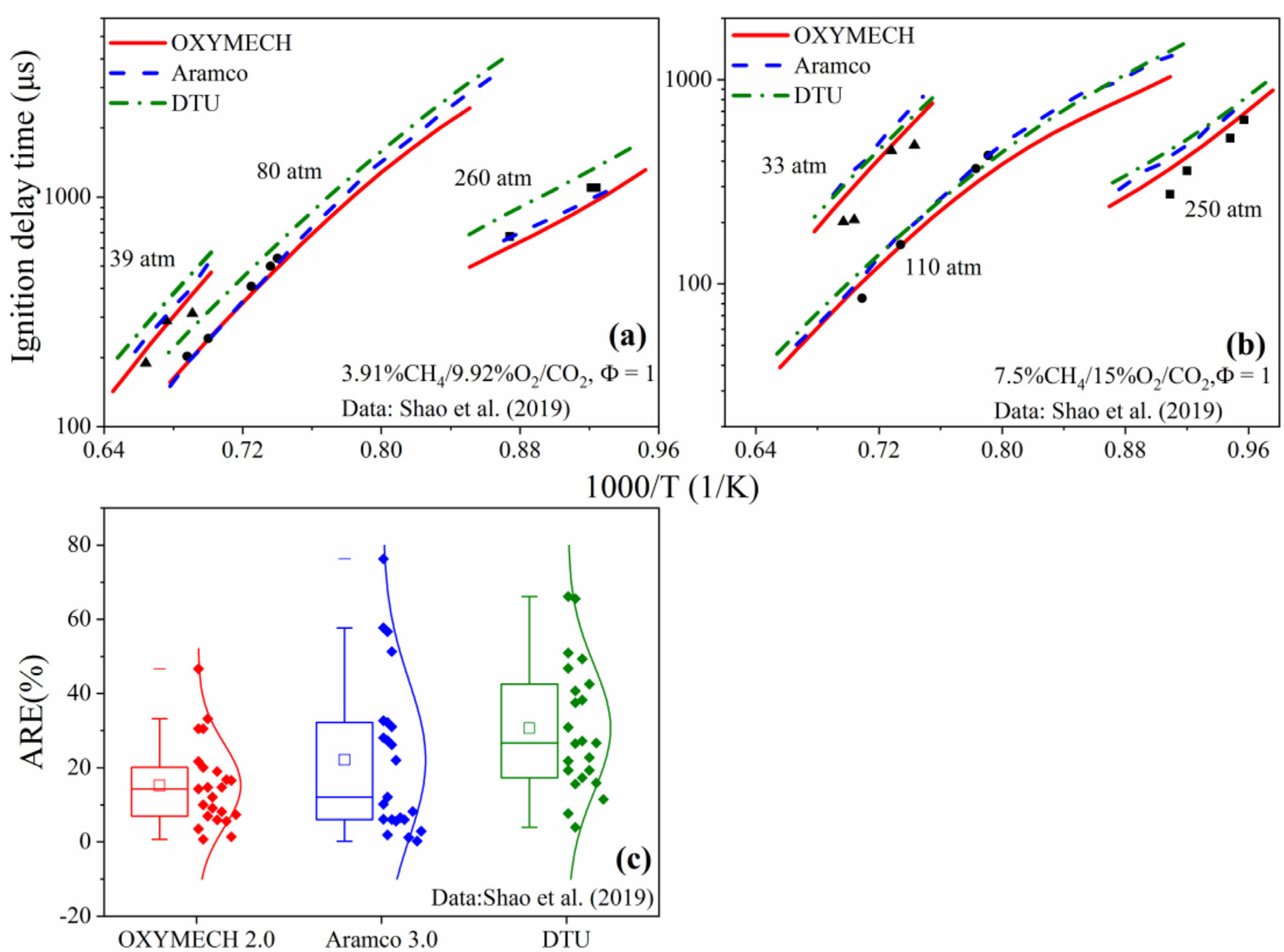

Fig. 8. Comparison of the IDTs of CH_4 diluted in CO_2 at high pressures obtained with the OXYMECH 2.0, Aramco 3.0, and DTU models.

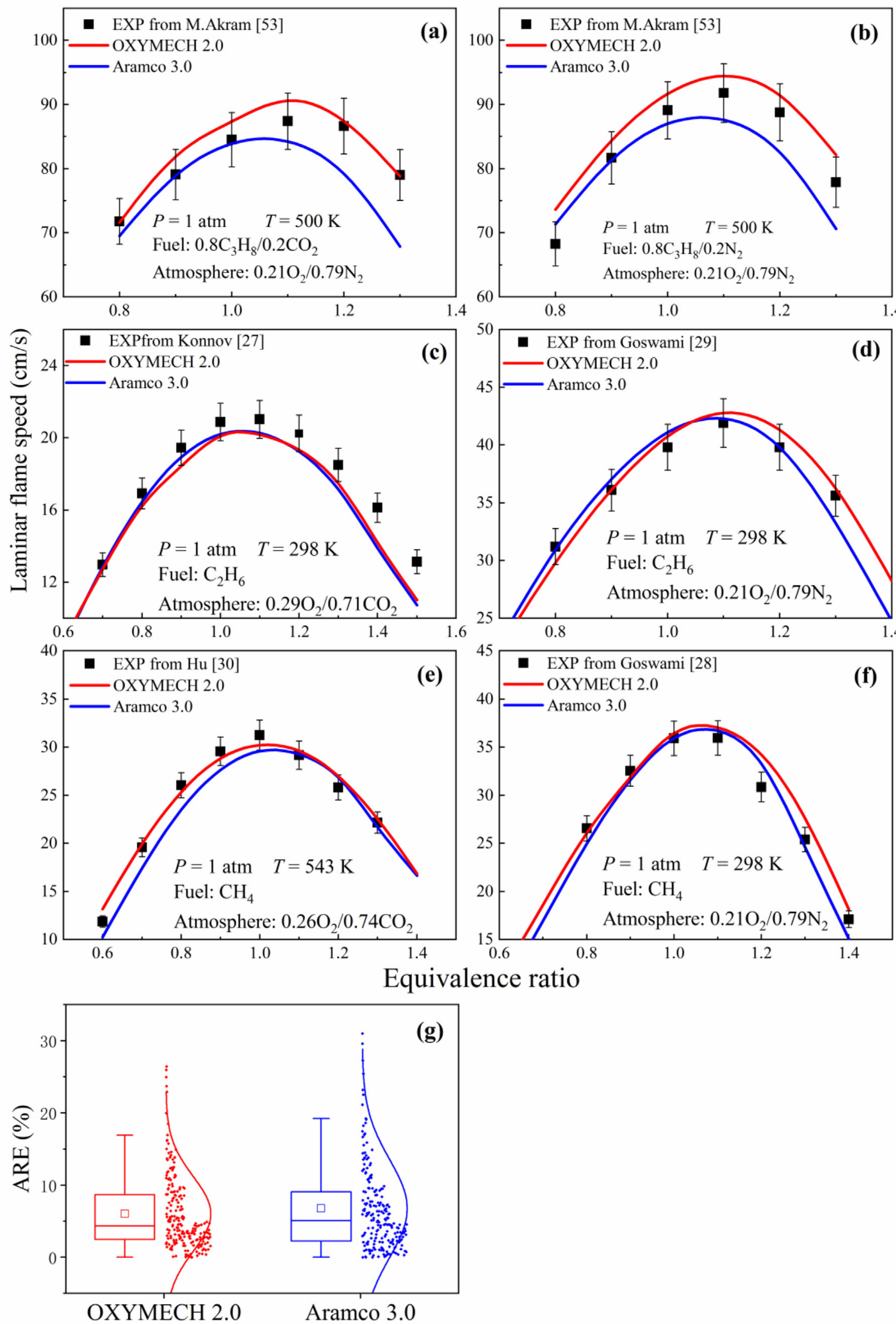

Fig. 9. Simulated and experimental laminar flame speeds.

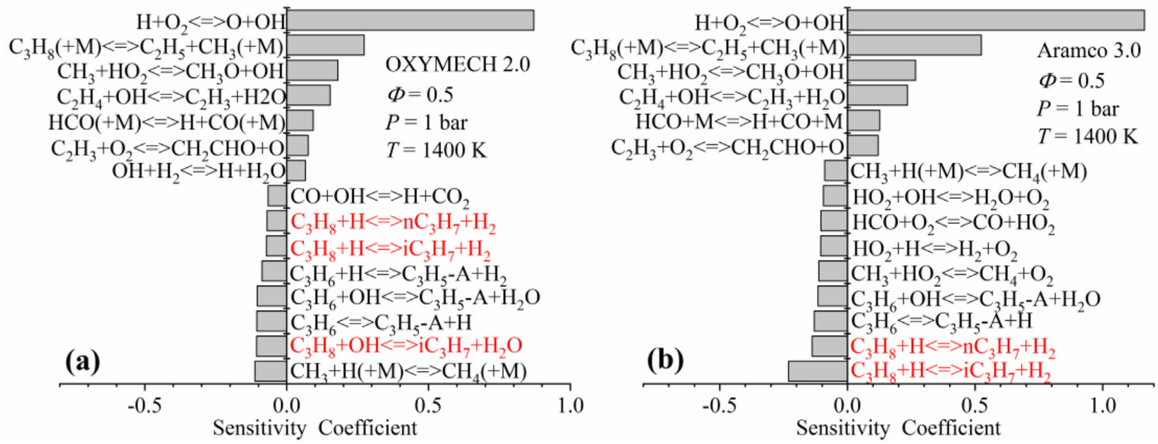

Fig. 10. Temperature sensitivity analysis at the ignition point for mix-1 at P = 1 bar and T = 1400 K using OXYMECH 2.0 and Aramco 3.0.

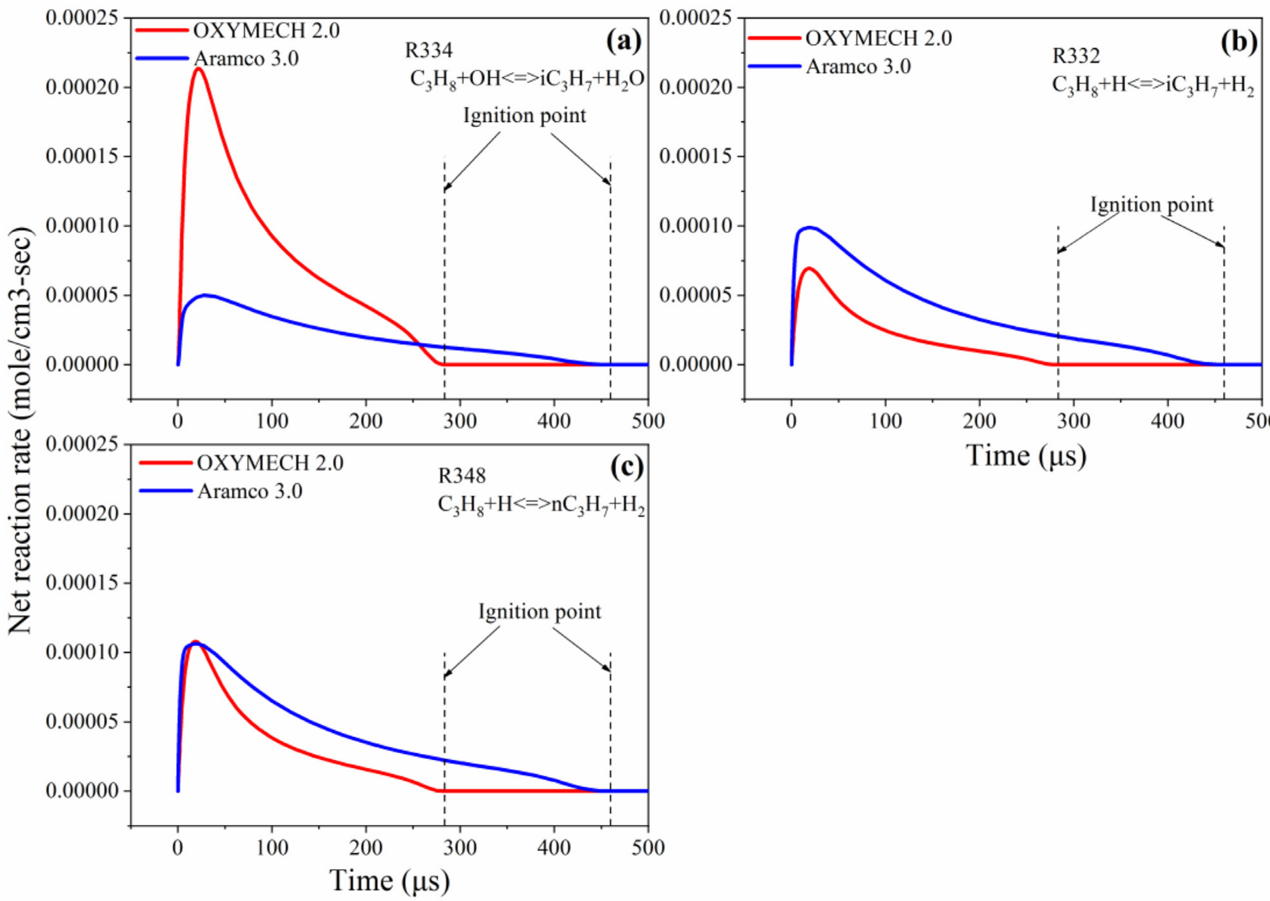

Fig. 11. Comparison of the net reaction rates of (a) R334, (b) R332, and (c) R348 in the OXYMECH 2.0 and Aramco 3.0 models at P = 1 bar and T = 1400 K.

At P = 10 bar, the increased pressure promotes pressure-dependent reactions. Figure 15 shows the top 20 reactions with high sensitivity coefficients for propane under conditions of 60% CO_2 at T = 1250 K and P = 10 bar. As shown in Fig. 15, with the increase in the equivalence ratio from 0.5 to 1.0 and 1.5, reaction C_3H_8 + H <=> nC_3H_7 + H_2 (R348) increases from 8th to 3rd and 2nd in importance, respectively, among reactions with negative sensitivity coefficients, while reaction C_3H_8 + OH <=> nC_3H_7 + H_2O (R350) increases from 7th to 4th and 3rd in importance, respectively. Reaction R350 has no significant effect on the IDT at P = 1 bar. Moreover, Fig. 16 displays the reaction pathway of propane at 20% fuel consumption, P = 10 bar, and T = 1250 K. It can be seen from Fig. 16 that the proportion of propane dehydrogenated through reaction R348 increases from 7.6% to 12.6% with the increase in the equivalence ratio from 0.5 to 1.5, which prolongs the IDT under fuel-rich conditions compared with the IDT under fuel-lean conditions. On the other hand, the proportion of propane dehydrogenated through reaction R350 decreases from 27.1% to 22.2% with the increase in equivalence ratio from 0.5 to 1.5, which shortens the IDT under fuel-rich conditions compared with that under fuel-lean conditions. Moreover, reaction C_3H_6 + OH <=> C_3H_5-A + H_2O (R499) is 2nd, 2nd, and 4th in importance among reactions with negative sensitivity coefficients at equivalence ratios of 0.5, 1.0, and 1.5, respectively. In Fig. 16,

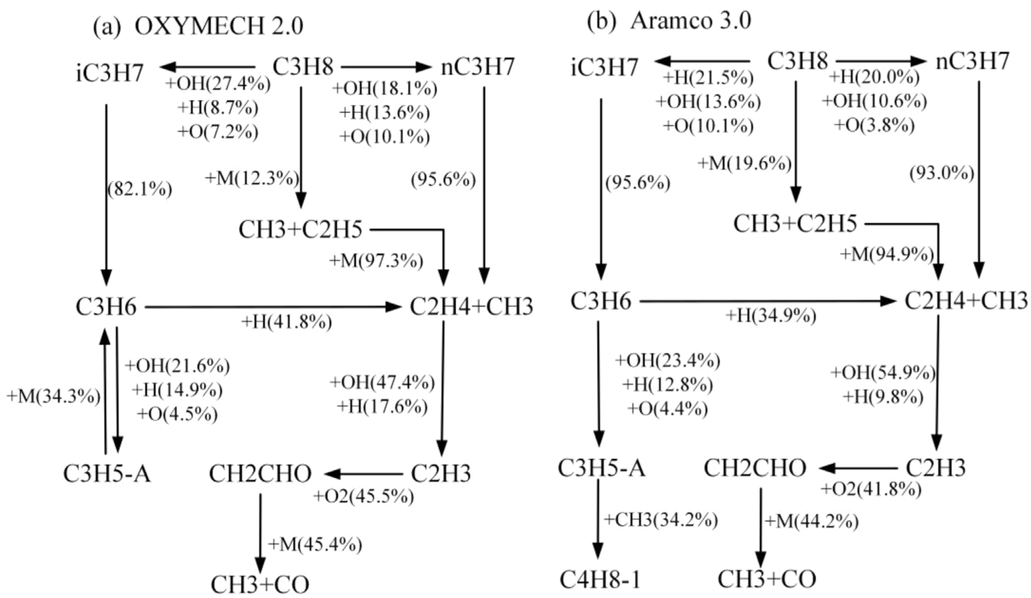

Fig. 12. Reaction pathways at 20% fuel consumption for mix-1 at P = 1 bar and T = 1400 K in the OXYMECH 2.0 and Aramco 3.0 models.

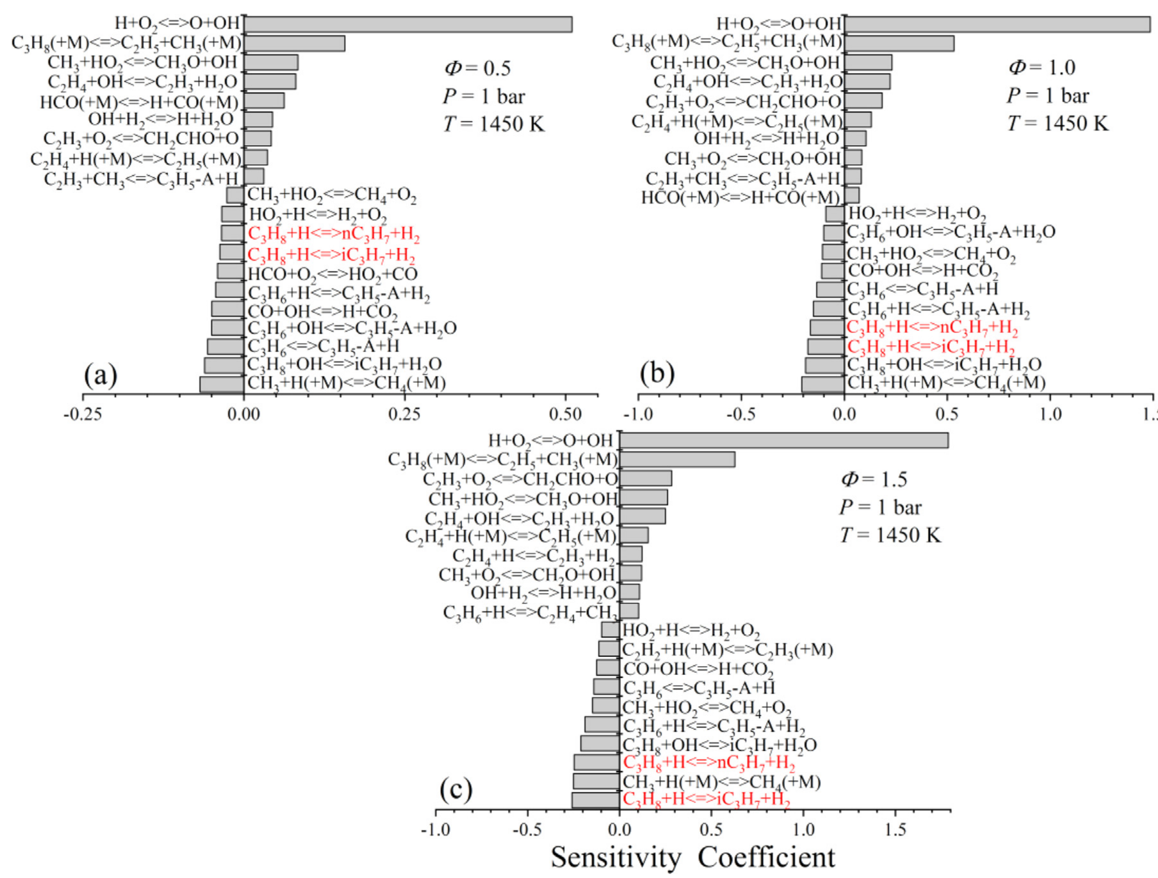

Fig. 13. Temperature sensitivity analysis at the ignition point using OXYMECH 2.0 at P = 1 bar and T = 1450 K.

the proportion of C_3H_6 dehydrogenated through reaction R499 decreases from 34.7% to 24.6%, which also shortens the IDT under fuel-rich conditions compared with that under fuel-lean conditions. Evidently, at P = 10 bar, as the equivalence ratio increases from 0.5 to 1.5, the suppression effects of reaction R348 compensate for the promotion effects of R350 and R499. Therefore, the differences in the IDTs at the three different equivalence ratios are not obvious at 10 bar and 1250 K.

3.5. Effects of CO_2 addition on the C_3H_8 ignition

Figure 2(f) shows that CO_2 inhibits the ignition of propane compared with N_2. To evaluate the effects of the physical and chemical properties of CO_2 on the ignition of propane diluted in CO_2, it is essential to separate the physical properties and chemical properties during the simulation. To this end, two artificial species, X and Y, were incorporated into the OXYMECH 2.0 model. Species X was

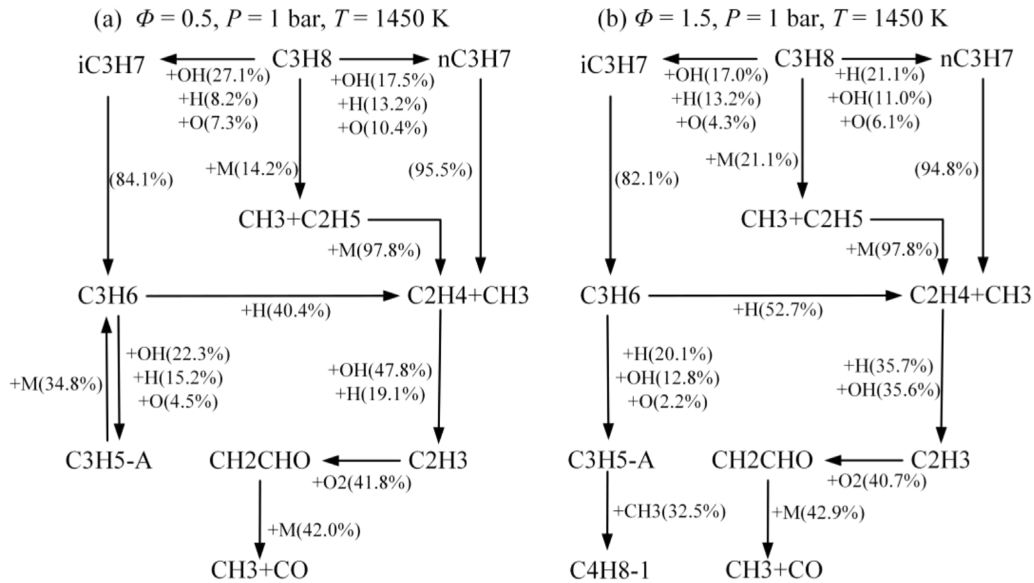

Fig. 14. Reaction pathways at 20% fuel consumption for mix-1 and mix-3 at P = 1 bar and T = 1450 K.

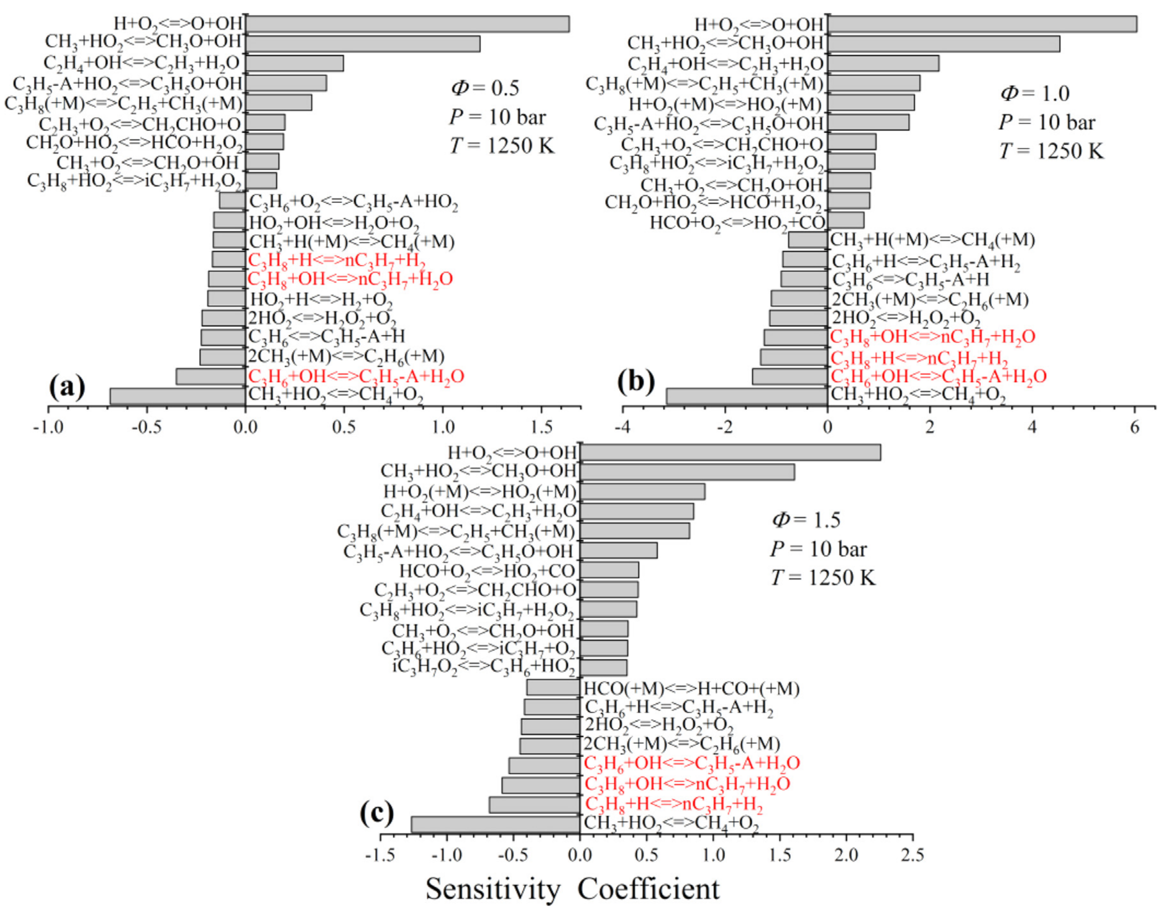

Fig. 15. Temperature sensitivity analysis at the ignition point using OXYMECH 2.0 at P = 10 bar and T = 1250 K.

assumed to have the same thermochemical properties as CO_2, but to be chemically inert. Thus, the differences in the propane IDTs between the O_2/X atmosphere (τ_X) and O_2/N_2 atmosphere (τ_{N_2}) demonstrate the effects of the physical properties of CO_2 on the ignition, while the differences in the IDTs between the O_2/X atmosphere (τ_X) and O_2/CO_2 atmosphere (τ_{CO_2}) represent the effects of the chemical properties of CO_2. The percent variation (PV) [54] of the effects was introduced to investigate the physical and chemical effects of CO_2 quantitatively. The PV of the physical effects was defined as follows:

$$PV_P = \frac{\tau_X - \tau_{N_2}}{\tau_X}. \tag{3}$$

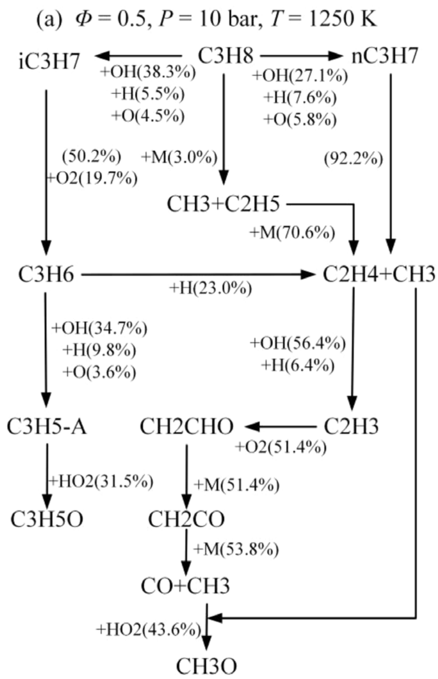

Fig. 16. Reaction pathways at 20% fuel consumption for mix-1 and mix-3 at P = 10 bar and T = 1250 K.

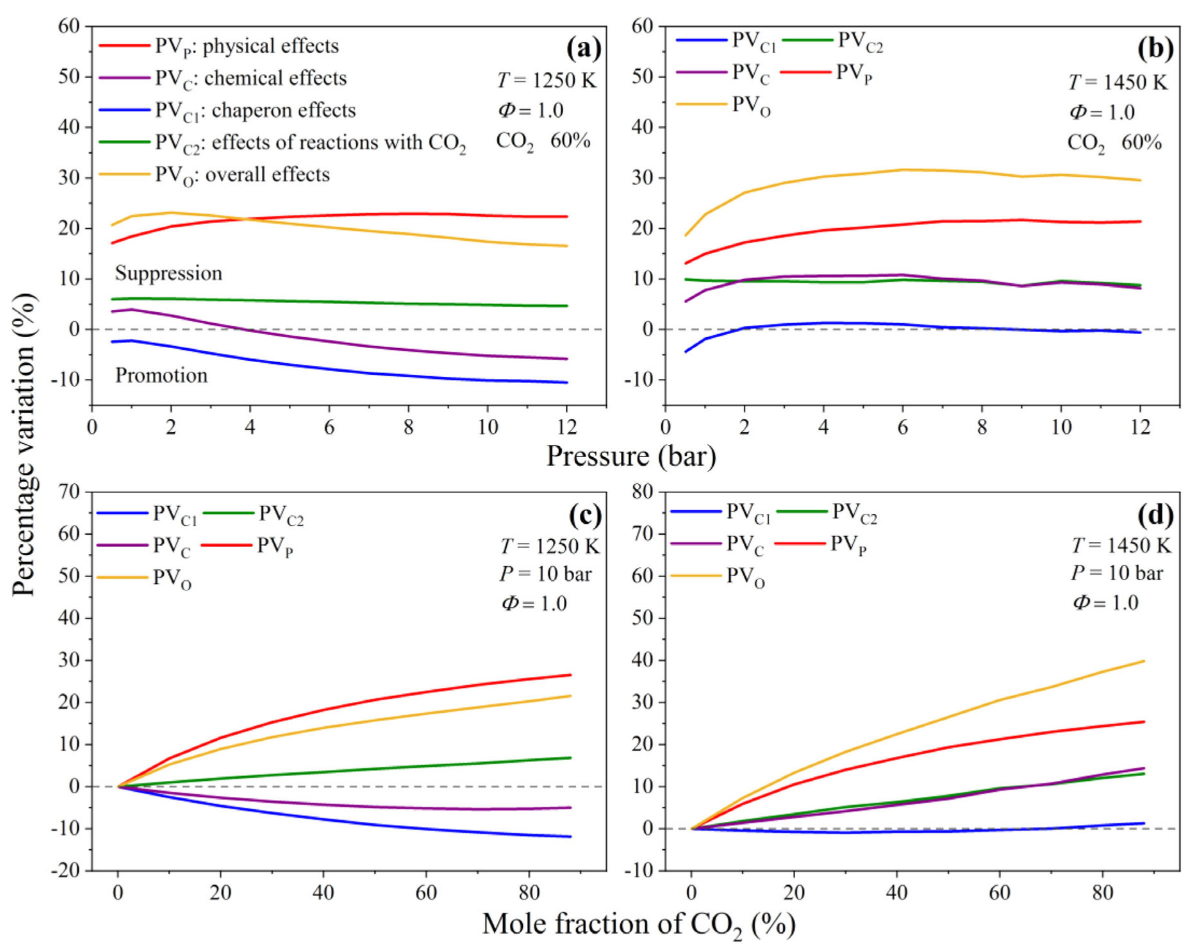

Fig. 17. Chemical and physical effects of CO_2 addition on C_3H_8 ignition. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The PV of the chemical effects was defined as follows:

$$PV_C = \frac{\tau_{CO_2} - \tau_X}{\tau_X}. \quad (4)$$

The sum of PV_C and PV_P represents the PV of the overall effects of CO_2 (PV_O). Moreover, species Y was assumed to have the same thermochemical properties and third-body collision efficiencies as CO_2; however, it did not participate in any chemical reactions. Thus, the differences in the IDTs between the O_2/X atmosphere (τ_X) and O_2/Y atmosphere (τ_Y) represent the chaperon effects of CO_2, while the differences in the IDTs between the O_2/Y atmosphere (τ_Y) and O_2/CO_2 atmosphere (τ_{CO_2}) represent the effects of the reactions with CO_2. The PV of the chaperon effects was defined as

$$PV_{C1} = \frac{\tau_Y - \tau_X}{\tau_X}. \quad (5)$$

The PV of the effects of the reactions with CO_2 was defined as

$$PV_{C2} = \frac{\tau_{CO_2} - \tau_Y}{\tau_X}. \quad (6)$$

Figure 17(a) and (b) show the dependence of the PV on the pressure for mix-2 at $T = 1250$ and 1450 K, while Fig. 17(c) and (d) show the dependence of the PV on the concentration of CO_2 at a pressure of 10 bar and equivalence ratio of 1.0. As shown in Fig. 17(a) and (b), the physical influence of CO_2 (marked by the red line) is much stronger than the chemical influence (marked by the purple line) due to the significantly higher volume specific heat of CO_2 compared to N_2. Thus, the physical influence of CO_2 is the reason for the longer IDTs in the CO_2 cases compared with those in the N_2 cases. At $T = 1250$ K, the effects of reactions with CO_2 (e.g., CO + OH <=> CO_2 + H, marked by the green line) suppress the ignition, and these change little with increased pressure, while the chaperon effects (e.g., C_3H_8 (+M) <=> C_2H_5 + CH3 (+M) and H + O_2 (+M) <=> HO_2 (+M), marked by the blue line) promote the ignition and increase with increased pressure. Therefore, the overall effects of CO_2 (marked by the yellow line) decrease gradually with the increase in pressure. At $T = 1450$ K, both the physical and chemical influences of CO_2 suppress the ignition, which is the reason for the longer IDTs in the CO_2 cases compared with those in the N_2 cases. The effects of reaction with CO_2 do not vary considerably with the increase in pressure, while the chaperon effects are very weak at $P > 2$ bar. Therefore, the total effects of CO_2 remain almost the same at $P > 2$ bar. For propane oxy-fuel combustion at the stoichiometric ratio, the effects of CO_2 on the ignition have a low sensitivity to the pressure. As shown in Fig. 17(c) and (d), the effects of CO_2 increase gradually when the concentration of CO_2 is greater than 30%; hence, there the IDTs do not vary significantly when the concentration of CO_2 increases from 60% to 80%.

4. Conclusion

The IDTs of propane under O_2/CO_2 atmospheres were measured in a shock tube. The experimental results show that the IDTs of propane increase with the equivalence ratio, and the effect of the equivalence ratio decreases with the increase in pressure from 1 to 10 bar. When the concentration of CO_2 is increased from 60% to 80%, the IDTs exhibit no significant change at $P = 10$ bar. An updated model (OXYMECH 2.0) for pressurized oxy-fuel combustion was proposed based on our previous model (OXYMECH 1.0). The OXYMECH 2.0, OXYMECH 1.0, Aramco 3.0, DTU, and CRECK models were evaluated in terms of the IDTs and LFSs of methane, ethane, and propane. The results indicated that OXYMECH 2.0 shows good performance for the prediction of C_1–C_3 IDTs in the pressure range from 1 atm to 250 atm and C_1–C_3 LFSs in the pressure range from 1 atm to 5 atm diluted in CO_2/N_2/Ar. However, further evaluation using available data from rapid compression machines and flow reactors is needed. The underestimation of the reaction rates of C_3H_8 + OH <=> iC_3H_7 + H_2O and the overestimation of the reaction rates of C_3H_8 + H <=> iC_3H_7 + H_2 and C_3H_8 + H <=> nC_3H_7 + H_2 in Aramco 3.0 are the reasons for the over-prediction of the IDTs of propane under O_2/CO_2 atmospheres by that model.

At 1 bar, as the equivalence ratio increases from the fuel-lean side to the fuel-rich side, fuel decomposition reactions become more important. The H-abstraction reactions of C_3H_8 by H radicals are the main reason for the increase in IDTs with increasing equivalence ratios. At 10 bar, the suppression effects of reaction R348 compensate for the promotion effects of R350 and R499. As a result, the differences in the IDTs between the three equivalence ratios are not obvious at 10 bar and 1250 K. Both the physical and chemical influences of CO_2 suppress the ignition, which is the reason for the longer IDTs in the CO_2 cases compared with those in the N_2 cases. For propane oxy-fuel combustion at the stoichiometric ratio, the effects of CO_2 on the ignition have low sensitivity to the pressure.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.06.024.

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